WiL update, 7 November 2026

Dataset decision · The loop closer · A real non-visual baseline · Detector · Maps

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1. The dataset set is fixed: eleven recordings, everything else retired

Recording	Role	Layout
`realistic_000`	Exp. 1, 3; Exp. 4 probe	U static
`realistic_001`	Exp. 1, 3; Exp. 4 probe	I static
`realistic_002`	Exp. 1, 3	U traffic
`realistic_003`	Exp. 1, 3	I traffic
`realistic_map`	Exp. 4 build; Exp. 3 cost	U static
`realistic_01_map`	Exp. 4 build; Exp. 3 cost	I static
`map_tiny`	Exp. 4 build	tiny
`realv6_000`	Exp. 1, 3 (real)	Gensurv
`realv7_000–002`	Exp. 1, 3 (real)	Gensurv

- Retired on 6 November: the `allsensor` and `nofloortexture` campaign, `relogged_20260916`, the October filtered runs, `realv2–realv5`
- Their results went to `trash/output/`, logged, and are cited nowhere
- **Dynamic environment** is now a *condition* of a recording, not a factor: no two recordings share a route
- Filtered arms are blank in both domains until the repeat is run

Source: `trash/output/RETIRED_20261106.txt`; registry in chapter 3

1. Six of eight result slots now hold data

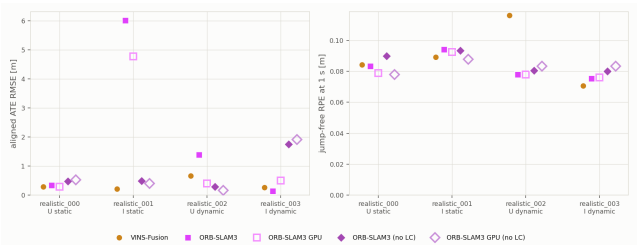
Experiment	Simulation	Real robot
1. Non-visual baseline	completed: 4 recordings, EKF at the VSLAM noise model	partial: wheel+gyro, fitted scale
2. YOLO detector	completed: 4 recordings, 4-class checkpoint	planned: no annotated frames
3. VIO + filtering	partial: 20 unfiltered runs + 18 cost repeats; no filtered run	partial: ORB-SLAM3 provisional; VINS no valid run
4. Dense map	partial: tiny passes, 4 realistic fail; large failed	planned

Changed since 22 October: Experiment 2 moved from untested to completed, Experiment 1 gained its real-robot half, and Experiment 3 gained both its cost measurement and a mechanism for ORB-SLAM3's error.

Source: chapter_4_experiments.tex, table 4.1

2. Twenty unfiltered runs: VINS is steady, ORB-SLAM3 is bimodal

- Factors: estimator, front end (CPU / GPU), loop closer (on / off). One run per cell, replay rate 1.0
- VINS-Fusion 0.22–0.66 m, no jump, on every recording
- ORB-SLAM3 0.14–6.02 m: eleven runs of sixteen between 0.14 and 0.52 m, five diverge
- Neither front end ever loses tracking: zero loss events, zero resets in twenty runs



Source: `output/compare/simulation/exp-realistic-ab/runs_metrics.csv`

2. Every ORB-SLAM3 divergence is the loop closer, in both directions

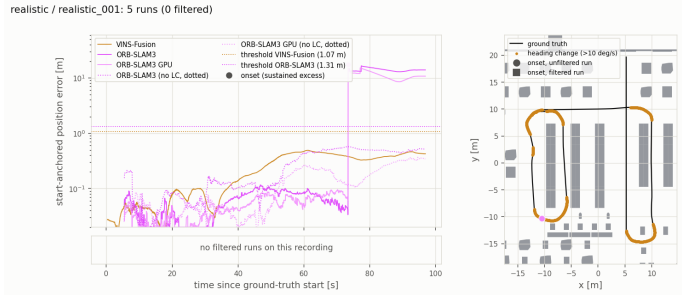
Recording	Layout	Closer on	Closer off	What the closer did
`realistic_000`	U static	0.34 / 0.29	0.48 / 0.52	two closures of 0.07–0.12 m: inert
`realistic_001`	I static	6.02 / 4.78	0.50 / 0.40	one false closure of 14 m
`realistic_002`	U traffic	1.39 / 0.40	0.29 / 0.17	one closure of 0.26 m; the 1.39 is an init offset
`realistic_003`	I traffic	0.14 / 0.50	1.76 / 1.91	one closure removes 5 m of vertical drift

Each cell is aligned ATE in metres, CPU front end / GPU front end. The closer state is read from each run's own log, never from its name.

Source: `loop_closing.csv` per run; `exp-realistic-ab`

2. The I layout's repeated aisles produce a false 14 m closure

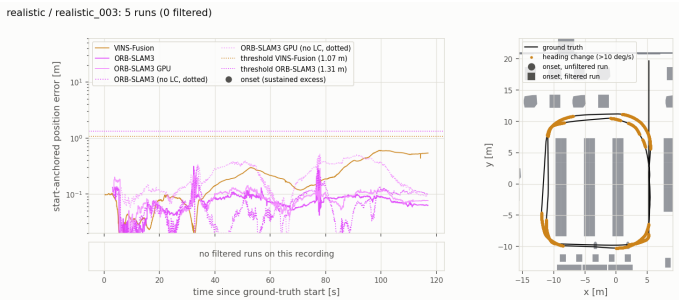
- Both front ends accept **one** closure at 73.2 s, 76 % along the route
- Correction 14.7 m (CPU) and 13.3 m (GPU)
- The estimate had drifted about 0.5 m before it: no real revisit needs a 14 m shift
- Matched keyframes are **one aisle apart**
- Four later candidates were rejected by the inertial geometry check, after the damage
- Both front ends failing at the same frame points at place recognition, not the extractor



Source: `timeline_realistic__realistic_001.png`; **step at 73 s**

2. On the same layout with traffic, only a closure removes the vertical drift

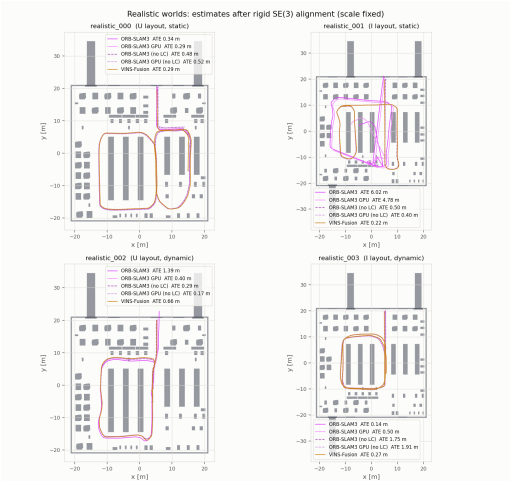
- Closure-free runs read 1.76 and 1.91 m ATE, yet their *planar* error never exceeds 0.7 m
- The difference is vertical: they climb to 4.7 and 5.1 m above the floor
- The loop-closing runs stay within 1.4 m and read 0.14 and 0.50 m
- Our ATE is three-dimensional, so it charges that climb to a floor robot: the planar onset table is the complement



Source: No run crosses its planar threshold on this recording

2. The GPU front end changes how many frames survive, not accuracy

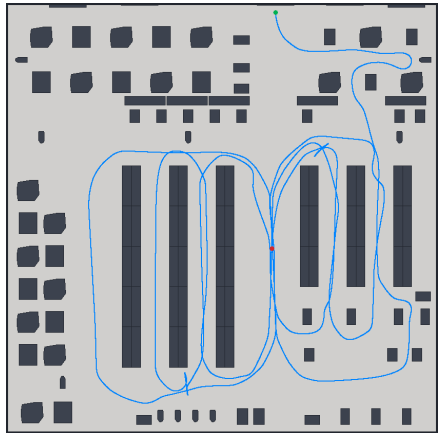
- GPU extractor keeps 97–99 % of each bag's frames; CPU keeps 72–94 %
- Tracking median 30–37 ms against 49–55 ms
- Across the eight CPU/GPU pairs at the same closer state, five differ by at most 0.16 m
- The three larger differences are each one run's discrete event, and go in both directions
- **Consequence**: at this frame budget the extractor is not the limiting stage of accuracy



Source: Solid = closer active, dashed = disabled, lighter = GPU front end

3. Cost is measured on the two mapping runs, three repeats each

- The four short recordings cannot support an uncertainty: one run per cell
- The mapping runs are 288 and 315 s over 374 and 385 m, every aisle entered
- They are also the only recordings that reach the project's 1.5 m s^{-1}: median 1.49 and 1.48 m s^{-1}
- Eighteen runs: ORB-SLAM3 CPU and GPU closure-free, and VINS-Fusion, three repeats of each



`realistic_map`: the U-layout mapping route

Source: `exp3-rules/mapping_resources.csv`, 18 runs

3. Neither pipeline is cheaper: it is a memory-against-cores trade

Recording	Pipeline	Front [ms]	Back [ms]	Kept	CPU [%]	RSS [MiB]
`realistic_map`	ORB-SLAM3 CPU	42.0	211.3	93	379	1732
	ORB-SLAM3 GPU	36.7	284.2	96	332	1794
	VINS-Fusion	38.1	44.3	91	819	479
`realistic_01_map`	ORB-SLAM3 CPU	43.9	220.4	91	381	1816
	ORB-SLAM3 GPU	51.9	405.7	75	361	1719
	VINS-Fusion	38.6	44.5	90	814	497

Mean over three repeats. VINS-Fusion optimises five to nine times faster than ORB-SLAM3's local mapping and holds its state in a quarter of the memory, for more than twice the processor time: about eight cores of one against three and a half. Front ends are within 6 ms.

Source: Mean of three repeats per configuration

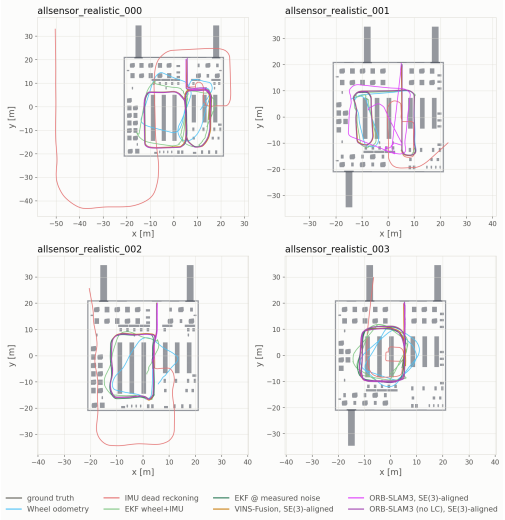
3. The I-layout front-end comparison is confounded with heat

- On the U recording the GPU front end behaves as the single runs showed: 37 against 42 ms, 96 against 93 % kept
- On the I recording the sign reverses: 52 against 44 ms, 75 against 91 % kept
- **But** each configuration was run as its own block: six GPU runs 21:04–22:04, six CPU runs 23:27–00:18, six VINS runs the next midday, on a laptop that throttles at 90 °C
- VINS-Fusion reads the *same* on both recordings (38.1 and 38.6 ms front end), so the recordings themselves do not impose a difference of that size
- Quote the U figures until the three configurations are repeated interleaved from a cool start

Source: Run start times from each run's `run_metadata.csv`

4. In simulation the EKF loses to VINS once both share a noise model

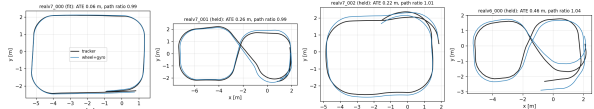
- Design decision: the EKF baseline now runs on the inertial noise the *visual* configurations declare, not its own measured model
- Reason: the two families must share one hand-set model before their accuracies compare at all
- At that model the EKF holds 0.68–2.80 m and loses to VINS-Fusion (0.22–0.66 m) on all four
- The same filter on its measured model holds 0.21–0.58 m and would beat VINS on three of four
- **So the whole gap is the noise model, none of it the architecture**



Source: Configured gyroscope density is 5–7× the measured value

4. The real robot now has a non-visual baseline, and vision does not win

- The rig publishes no metric wheel speed and no joint states; its steering feedback is a two-valued command echo, so no bicycle model and no EKF
- What works: speed feedback $\times$ one scale k, heading from the gyroscope with its rest bias removed
- k, the gyro *axis* and its sign are fitted on `realv7_000` only, then held
- The yaw rate is on the sensor's **y** axis; its z axis integrates to zero over a 6.3 rad heading sweep



Recording	Role	Wheel+gyro	Best ORB-SLAM3
`realv7_000`	fit	0.06	0.25
`realv7_001`	held	0.26	0.44
`realv7_002`	held	0.22	0.47
`realv6_000`	held	0.46	0.46

Source: Planar ATE in metres; lever arm zero on both sides

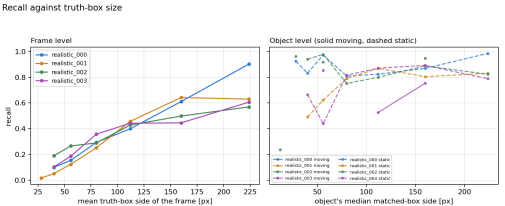
4. What bounds that comparison

- One tuned constant. k is fitted against the same tracker the baseline is scored on, albeit on a different recording. The fit recording's 0.06 m is a calibration, not a result
- The fitted $k = 0.325$ agrees with the plain path-length ratio of each `realv7` bag (0.323–0.329), which is a weak independent check
- Path ratios come out within 4 % of one, where the visual runs read 5–15 % long
- Routes are 22–37 m at 0.2 m s^{-1}; gyroscope drift grows with route length and vision's need not
- The marker-to-IMU lever arm is unmeasured on both sides
- **Fix**: a measured scale from wheel circumference and gear ratio, and the steering angle recorded as a measurement, which would also restore the EKF with the 0.46 m wheelbase

Source: `script/real_wheel_baseline.py`, `output/compare/real/exp1-wheel/`

5. The four-class checkpoint makes the pallet jacks scorable

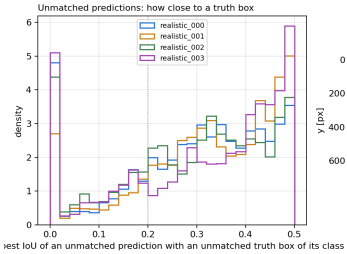
- `weight/runs_1/.../best.pt`: bin, box, bucket, **forklift**; the fourth fires on the pallet jacks, which are the moving objects
- Scored on all four realistic recordings, 12 698 frames
- Pooled precision **0.747**; class F1 **0.994** given a match
- Recall **0.502** over the boxes the exemptions keep, **0.288** over everything in view
- Moving-object recall **0.288** — the ceiling on any filtering benefit downstream
- The walking people are simulator actors: no pose, no mesh, outside the class set, so unscorable *and* unmaskable

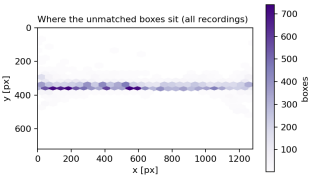


Source: `output/yolo_eval/*_4class/`; confidence 0.35, matching IoU 0.5

5. The miss is a size effect; the false positives are near misses

- Frame-level recall rises monotonically with truth-box side: 0.02 at 24–32 px to 0.64 above 192 px
- The moving pallet jacks lie on the *same* curve as the parked ones, so 0.29 is a distance effect, not a motion effect
- Of the unmatched predictions, 74–79 % overlap the right object below the IoU threshold; only 8–9 % land on nothing
- A filter that masks by box does not care which
- Misses come as flicker: 39–47 % of gaps are a single frame





Source: `output/compare/simulation/exp2-insights/`

6. All four realistic maps fail the criterion fixed before scoring

Database	Layout, build	Precision	Recall seen	False obstacles	Drivable
`map_realistic`	U, October	81.2	82.5	3.84	96.1
`..._realistic_map`	U, November	77.8	74.9	5.00	94.9
`map_realistic_01`	I, October	84.8	81.4	4.39	94.9
`..._realistic_01_map`	I, November	82.2	77.1	10.19	89.4
Criterion	fixed before scoring	≥ 90	≥ 85	≤ 1	—

All values in percent, probed with a recording that did not build the map. Two configurations × two layouts, one build each, so the parameter differences below are signatures rather than ablations.

Source: `exp4-realistic-maps/realistic_map_metrics.csv`

6. The November configuration is worse, through three parameters

- The November builds differ from the October ones in exactly three RTAB-Map parameters, and each leaves its own signature:
  - lower detection rate → fewer nodes, fewer views per cell
  - longer grid range → phantom cells along the west wall
  - higher ground-height cut → obstacles in the aisles, and on the I layout the blocking phantoms appear *mid-route* (median first node 27, against 1 everywhere else)
- The false obstacles are **not** thickened shelving: they are phantoms laid during the inertial initialisation transient at the spawn, plus wall bands displaced about 0.5 m
- **The correction the data predict**: discard the cells of the first few nodes, or start mapping after the transient settles
- The three parameters changed together, so which one dominates is untested

Source: `script/exp4_insights.py -realistic`, Insight 6

7. The report is rebuilt on the fixed dataset set

- **Chapter 3** purged of the retired campaign: nine fragments of the old report dropped and rewritten from current measurements — registry, worlds, sensor fidelity, the four experiments' methods, status and deviations, real rig
- Every number in it re-measured on the recordings in use: wheel slip under 0.4 %, gyroscope noise 5–7× below the configured value, the inertial orientation field confirmed to be ground truth
- **Chapter 4** slots 1–3 are now generated from the insight notes by `script/build_chapter4.py`; edit the note, regenerate
- **Chapter 5** rewritten; the two hypotheses the project is named for still have no real-robot evidence
- 129 pages, no undefined references, four overfull lines, all pre-existing

Source: `script/build_final_report.sh`

7. What is open, in the order it should be closed

1. **Give Experiment 3 its test.** Both pipelines filtered and unfiltered on `realistic_002` and `_003`, paired under the admissibility gates, with the closer *disabled* on the I layout so the false closure does not dominate both arms
2. **Bound the loop closer.** A maximum correction against the inertial drift bound, tested on `realistic_001` where the 14 m closure is reproducible in both front ends
3. **Repeat the cost runs interleaved** from a cool start, so the I-layout front-end comparison is readable
4. **Real robot**: exercise the fork's initialisation switches, re-run VINS with the corrected gravity norm, measure the marker-to-IMU lever arm, measure the wheel-speed scale
5. **Detector**: a person class with projectable geometry, or an annotated frame subset, so the agents the filter exists for are scorable
6. **Maps**: start mapping after the inertial transient, and reconsider the November range and ground-height parameters

Summary

- The report is now built from eleven recordings and nothing else; everything retired is logged, not deleted
- **ORB-SLAM3's error in simulation is its loop closer**, in both directions: a false 14 m closure on the I layout's repeated aisles, and a vertical drift on the same layout that only a closure removes. The front end changes retention, not accuracy
- **VINS-Fusion** is the steadier estimator in simulation and costs a quarter of the memory for twice the cores, but still has no valid run on the real robot
- **Vision has not earned its place over dead reckoning** in either domain, on routes of this length
- **The detector finds under a third of the moving objects**, which bounds any filtering benefit before the filter is tested
- **All four realistic maps fail** the map-quality criterion, and the cause is the initialisation transient, not the shelving